

A candidate counterexample to Geodesic cycles and Tutte's Theorem

github.com/graph-theory-AI — machine-generated candidate writeup

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Abstract

The eight-vertex graph obtained by inserting a degree-three vertex into every face of a tetrahedron is a counterexample. This candidate disproof was generated by GPT-6 Astra in a single model pass for the Graph Theory AI project. It was selected from the campaign because the model marked `it would_publish: true`; it has not been checked by the adversarial referee, a human mathematician, or a novelty search.

1 The problem and the claimed result

This document concerns [Geodesic cycles and Tutte's Theorem](http://www.openproblemgarden.org/op/geodesic_cycles_and_tuttes_theorem) from http://www.openproblemgarden.org/op/geodesic_cycles_and_tuttes_theorem.

1.1 Problem statement

Problem If G is a 3 -connected finite graph, is there an assignment of lengths $\ell : E(G) \rightarrow \mathbb{R}^+$ to the edges of G , such that every ℓ -geodesic cycle is peripheral ?

1.2 Catalog context

A cycle C is ℓ -geodesic if for every two vertices x, y on C there is no $x - y$ ~path in G shorter, with respect to ℓ , than both $x - y$ ~arcs on C . It is not hard to prove [GS] that for every finite graph G and every assignment of edge lengths $\ell : E(G) \rightarrow \mathbb{R}^+$ the ℓ -geodesic cycles of G generate its cycle space. Thus, a positive answer to the problem would imply a new proof of Tutte's classical theorem [T] that the peripheral cycles of a 3 -connected finite graph generate its cycle space.

1.3 Claimed result

Model verdict: disproved (confidence: high).

The eight-vertex graph obtained by inserting a degree-three vertex into every face of a tetrahedron is a counterexample.

The model's own caveat: Peripheral has its standard induced, vertex-nonseparating meaning; bibliographical novelty has not been checked.

2 Model writeup

The text below is the model output verbatim, apart from moving its machine-readable verdict block into the summary above and shifting Markdown heading levels for this document. Mechanical TeX defects and equation tags were normalized where needed for compilation.

2.1 An eight-vertex counterexample

A cycle is **peripheral** if it is induced and deleting its vertices leaves a connected graph.

Let

$$B = \{b_1, b_2, b_3, b_4\}, \quad S = \{s_1, s_2, s_3, s_4\},$$

and define G by

$$V(G) = B \cup S, \quad E(G) = \{b_i b_j : i < j\} \cup \{s_i b_j : i \neq j\}.$$

Thus B induces a K_4 , the set S is independent, and s_i is adjacent to the three vertices of $B \setminus \{b_i\}$.

Equivalently, start with a tetrahedron embedded on the sphere and insert one vertex into each face, joining it to that face's three vertices.

Theorem. For every assignment $\ell : E(G) \rightarrow \mathbb{R}_{>0}$, some ℓ -geodesic cycle of G is not peripheral.

2.1.1 1. Connectivity and peripheral cycles of G

The graph G is 3-connected. Indeed, after deleting at most two vertices, at least two vertices of B remain and induce a connected clique. Every surviving s_i retains at least one of its three neighbours in B , so the remaining graph is connected.

There are eight vertices and eighteen edges, hence

$$\dim_{\mathbb{F}_2} \mathcal{C}(G) = 18 - 8 + 1 = 11.$$

We next classify the peripheral cycles.

Every induced cycle of G is a triangle. To see this, an induced cycle of length at least four cannot contain s_i : its two neighbours along the cycle would belong to B , and the edge between them would be a chord. A cycle of length at least four entirely within B also has a chord.

The triangles are of two types:

- The four triangles

$$T_i = G[B \setminus \{b_i\}]$$

are **not peripheral**. In $G - V(T_i)$, the vertex s_i is isolated, whereas the other remaining vertices form a nonempty connected component.

- The twelve triangles

$$P_{i;jk} = s_i b_j b_k s_i, \quad j < k, \quad i \notin \{j, k\},$$

are **peripheral**. After deleting such a triangle, two adjacent vertices of B remain. Every other surviving vertex of S has a neighbour among these two vertices.

Consequently, G has exactly twelve peripheral cycles, all triangles. Each contains **exactly one edge of the core** K_4 .

2.1.2 2. Two metric facts

We use two elementary facts, including their proofs to handle arbitrary positive lengths and possible ties.

2.1.2.1 Fact 1: Geodesic cycles generate the cycle space Let H be any finite graph with positive edge lengths. If a cycle C is not geodesic, there is a path P between vertices $x, y \in V(C)$ with

$$\ell(P) < d_C(x, y),$$

where d_C is distance measured within C .

There is then a path Q , with distinct endpoints $u, v \in V(C)$ and all internal vertices outside C , that is shorter than both u - v arcs of C . Indeed, otherwise every excursion of P outside C could be replaced by a no-longer arc of C . This would produce an x - y walk in C of length at most $\ell(P) < d_C(x, y)$, a contradiction.

Write the two u - v arcs as A, B . The cycles

$$C_1 = A \cup Q, \quad C_2 = B \cup Q$$

satisfy

$$E(C) = E(C_1) \triangle E(C_2), \quad \ell(C_1) < \ell(C), \quad \ell(C_2) < \ell(C).$$

Induction through the finitely many cycle lengths proves that the geodesic cycles generate $\mathcal{C}(H)$.

2.1.2.2 Fact 2: An admissible assignment can be perturbed to have unique shortest paths

Suppose an assignment has the property that every geodesic cycle is peripheral. For each nonperipheral cycle, choose one path witnessing its failure to be geodesic. Each witness gives two strict linear inequalities: the path is shorter than each of the two corresponding cycle arcs.

There are finitely many nonperipheral cycles. Thus these inequalities, and positivity of all edge lengths, persist throughout an open neighbourhood of the assignment.

Inside this neighbourhood, choose lengths avoiding all equalities

$$\ell(P) = \ell(Q)$$

between distinct simple paths with the same endpoints. These equalities form finitely many proper hyperplanes. The perturbed assignment still has no nonperipheral geodesic cycle, and every pair of vertices now has a unique shortest path.

Importantly, this argument preserves the strict shortcuts for nonperipheral cycles; it does **not** assume that particular peripheral cycles remain geodesic.

2.1.3 3. Contradiction

Suppose that G admits a positive length assignment for which every geodesic cycle is peripheral. By Fact 2, assume additionally that shortest paths between vertices are unique.

By Fact 1, the geodesic cycles generate the 11-dimensional cycle space. Since all of them belong to the twelve peripheral triangles listed above, at least eleven of those triangles must be geodesic. Therefore

$$\boxed{\text{At most one peripheral triangle is not geodesic.}} \quad (1)$$

Call an edge uv **tight** if

$$\ell(uv) = d_\ell(u, v).$$

Every edge satisfies $\ell(uv) \geq d_\ell(u, v)$, so a non-tight edge has strictly greater length than the distance between its endpoints.

Consider a core edge $e = b_j b_k$. Let $\{p, q\} = \{1, 2, 3, 4\} \setminus \{j, k\}$. Exactly two peripheral triangles contain e :

$$P_{p;jk} \quad \text{and} \quad P_{q;jk}.$$

If e is non-tight, these two triangles cannot both be geodesic. Otherwise, geodesicity for the pair b_j, b_k would imply

$$\ell(b_j s_p) + \ell(s_p b_k) = d_\ell(b_j, b_k) = \ell(b_j s_q) + \ell(s_q b_k).$$

The two distinct paths $b_j s_p b_k$ and $b_j s_q b_k$ would both be shortest, contradicting uniqueness.

Thus every non-tight core edge belongs to a non-geodesic peripheral triangle. Different core edges require different such triangles, because each peripheral triangle contains exactly one core edge. By (1),

$$\boxed{\text{At most one core edge is non-tight.}} \quad (2)$$

There is consequently a core triangle T_i containing no non-tight edge: if the exceptional edge exists, choose a core triangle avoiding it.

All three edges of this T_i are tight. For every pair of vertices of T_i , their joining edge therefore realizes their distance in G . Hence T_i is geodesic.

But T_i is not peripheral, as established in Section 1. This is the required contradiction. $\square$

2.1.4 Completeness and status

The perturbation argument covers assignments with tied shortest paths as well as generic assignments. Thus the counterexample rules out **every** positive edge-length assignment, not merely those in general position.

The graph is simple, planar, and 3-connected, so the assertion fails even within the planar subclass. The proof is self-contained; I have not checked whether this counterexample or argument has previously appeared in the literature.

A Provenance and review status

UNREFEREED MODEL OUTPUT. This document was selected solely because GPT-6 Astra labelled its own result `would_publish: true` and returned `proved` or `disproved`. It has not passed the adversarial LLM referee used for the earlier Sol campaign, has not been checked by a human mathematician, and has not been checked for novelty. Treat every mathematical and bibliographic claim as unverified.

Catalog id	geodesic_cycles_and_tuttes_theorem
Catalog entry	Geodesic cycles and Tutte's Theorem
Source corpus	opg
Campaign leg	OpenProblemGarden first pass (attacks_opg)
Source paper / entry	http://www.openproblemgarden.org/op/geodesic_cycles_and_tuttes_theorem
Model verdict	disproved (confidence: high)
Selection criterion	would_publish: true and verdict proved or disproved
Model	gpt-6-astra, reasoning effort max, mode=pro, flex service tier
Original artifact	attacks_opg/geodesic_cycles_and_tuttes_theorem/output.md
Independent review	None

Model's caveats

Peripheral has its standard induced,
vertex-nonseparating meaning; bibliographical
novelty has not been checked.

The generated Markdown and LaTeX sources for this PDF are stored under `to_review_astra/src/geodesic_cycles_and_t`